Timur Javid

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Education

University of Illinois Urbana-Champaign (UIUC) **August 2022 – Present**

Doctor of Philosophy in Electrical and Computer Engineering

Master of Computer Science (5-year BS/MCS) **August 2017 – May 2022**

Bachelor of Science in Computer Science *GPA: 3.90*

Skills and Interests

Languages: Python • C • C++ **Tools & Skills:** SciPy • NumPy • PennyLane • Qiskit • PyTorch • Embedded Dev.
Hobbies: Hiking & Backpacking • Photography • Pizza-making

Experience

Kwiat Quantum Information Group at UIUC **May 2021 – Present**

Graduate Research Assistant

- Integrated OpenCV detection and multi-threaded C++ code with PID control to improve drone-to-drone locking for quantum key distribution, decreasing average quantum channel loss during flight from 20.6 dB to 9.68 dB.
- Designed and implemented firmware on an STM32 micro-controller for entanglement source experiment to be tested on the International Space Station, in collaboration with the Laboratory for Advanced Space Systems at Illinois.
- Analyzed performance of variational quantum optimization of polarization qubits for mitigating fiber noise, setting up a quantum optical experiment and automating control of the experiment.
- Maintaining an open-source Python package (Quantum-Tomography) and webpage utilizing SciPy optimization for quantum state and process tomography of optical systems, deployed as a public webpage for use by experimentalists.

Aliro Quantum **May 2025 – Dec 2025**

Simulation Team Intern, Aliro Quantum Network Simulator

- Implemented quantum process tomography in Python using Aliro's quantum network simulator to serve as an example application for customers.
- Integrated the quantum network simulator with PennyLane, allowing for the use of PennyLane's quantum circuit and optimization framework for network optimization tasks.

IBM Research **May 2023 – August 2023**

Quantum Researcher Intern, Quantum State Transfer Systems

- Produced an experimental quantum optical set-up for ancilla-assisted quantum process tomography with spontaneous parametric downconversion (SPDC) waveguides and superconducting nanowire single photon detectors (SNSPD) in Bluefors dilution refrigerator.
- Cooperated with experimentalists and theorists on analysis of novel quantum state transfer systems.
- Analyzed novel quantum state transfer systems by implementing and executing quantum process tomography protocols, using Qiskit to gather data.
- Hands-on experience with fiber splicing, opto-mechanics, dilution refrigerator bring-up, optical alignment, and set-up and control of SPDC waveguides and SNSPDs.

IBM Research **May 2022 – August 2022**

Quantum Researcher Intern, Quantum Observability

- Incorporated three live data sources from quantum systems monitoring into a backend data pipeline utilizing Apache, Kafka, MQTT, and InfluxDB in Python, enabling observability for IBM Quantum systems.
- Enhanced data logging for existing alerts and notification system with a new data pipeline, bypassing logging to hard disk.
- Analyzed historical data to develop data visualizations with feedback from quantum researchers across teams.
- Constructed a front-end interface with the Carbon/React framework to display live data from quantum systems, linked with our InfluxDB backend.

Amazon Web Services (AWS) **May 2020 – August 2020**

Software Development Engineer Intern, IoT Edge Qualification

- Spearheaded and developed a front-end user interface for an existing hardware qualification tool using React and Redux and a Go backend within two months.
- Developed a web API in Go, allowing device tests to be run from HTTP requests and allowing users to obtain results from qualification tests.
- Integrated web API with a React/Redux front-end, providing users a visual representation of device tests, documentation for qualification standards, and feedback for which IoT services their hardware is compatible with.
- Led design meetings for the project, producing design documents and integrating feedback from code reviews and weekly meetings with peers and supervisors.